

12 System configuration controller (SYSCFG)

12.1 Introduction

The devices feature a set of configuration registers. The objectives of this section is to describe in details the system configuration controller.

12.2 SYSCFG main features

The system configuration controller main functions are the following:

- Analog switch configuration management
 - I2C Fm+ configuration
 - Selection of the Ethernet PHY interface.
 - Management of the external interrupt line connection to the GPIOs
 - Management of the I/O compensation cell
 - Getting readout protection and Flash memory bank swap informations
 - Management of boot sequences and boot addresses
 - Management BOR reset level
 - Management of Flash memory secured and protected sector status
 - Management Flash memory write protections status
 - Management of DTCM secured section status
 - Management of independent watchdog behavior (hardware or software / freeze)
 - Reset generation in Stop and Standby mode
 - Secure mode enabling/disabling status.
- On STM32H750xB devices, only the statuses of Bank 1 option bytes are applicable.

12.3 SYSCFG registers

12.3.1 SYSCFG peripheral mode configuration register (SYSCFG_PMCr)

Address offset: 0x04

Reset value: 0x0X00 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	PC3SO	PC2SO	PA1SO	PA0SO	EPIS[2:0]			Res.	Res.	Res.	Res.	Res.
				rw	rw	rw	rw	rw	rw	rw					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	BOOSTV DDSEL	BOOSTE	PB9 FMP	PB8 FMP	PB7 FMP	PB6 FMP	I2C4 FMP	I2C3 FMP	I2C2 FMP	I2C1 FMP
						rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bit 27 **PC3SO**: PC3 Switch Open

This bit controls the analog switch between PC3 and PC3_C (dual pad)
 0: Analog switch closed (pads are connected through the analog switch)
 1: Analog switch open (2 separated pads)

Bit 26 **PC2SO**: PC2 Switch Open

This bit controls the analog switch between PC2 and PC2_C (dual pad)
 0: Analog switch closed (pads are connected through the analog switch)
 1: Analog switch open (2 separated pads)

Bit 25 **PA1SO**: PA1 Switch Open

This bit controls the analog switch between PA1 and PA1_C (dual pad)
 0: Analog switch closed (pads are connected through the analog switch)
 1: Analog switch open (2 separated pads)

Bit 24 **PA0SO**: PA0 Switch Open

This bit controls the analog switch between PA0 and PA0_C (dual pad)
 0: Analog switch closed (pads are connected through the analog switch)
 1: Analog switch open (2 separated pads)

Bits 23:21 **EPIS[2:0]**: Ethernet PHY Interface Selection

These bits select the Ethernet PHY interface.

000: MII
 001: Reserved
 010: Reserved
 011: Reserved
 100: RMII
 101: Reserved
 110: Reserved
 111: Reserved

Bits 20:10 Reserved, must be kept at reset value.

Bit 9 **BOOSTVDDSEL**: Analog switch supply voltage selection (V_{DD} / V_{DDA} /booster)

To avoid current consumption due to booster activation when $V_{DDA} < 2.7$ V and $V_{DD} > 2.7$ V, V_{DD} can be selected as supply voltage for analog switches. In this case, the BOOSTE bit should be cleared to avoid unwanted power consumption.

When both $V_{DD} < 2.7$ V and $V_{DDA} < 2.7$ V, the booster is still needed to obtain full AC performances from I/O analog switches.

0: V_{DD} selected as analog switch supply voltage (when BOOSTE bit is cleared)
 1: V_{DDA} selected as analog switch supply voltage

Note: This bit is available only on devices revision X and higher.

Bit 8 **BOOSTE**: Booster Enable

This bit enables the booster to reduce the total harmonic distortion of the analog switch when the supply voltage is lower than 2.7 V.

Activating the booster allows to guaranty the analog switch AC performance when the supply voltage is below 2.7 V: in this case, the analog switch performance is the same on the full voltage range.

0: Booster disabled
 1: Booster enabled

- Bit 7 **PB9FMP**: PB(9) Fm+
This bit enables I2C Fm+ on PB(9).
0: Fm+ disabled
1: Fm+ enabled
- Bit 6 **PB8FMP**: PB(8) Fm+
This bit enables I2C Fm+ on PB(8).
0: Fm+ disabled
1: Fm+ enabled
- Bit 5 **PB7FMP**: PB(7) Fm+
this bit enables I2C Fm+ on PB(7).
0: Fm+ disabled
1: Fm+ enabled
- Bit 4 **PB6FMP**: PB(6) Fm+
This bit enables I2C Fm+ on PB(6).
0: Fm+ disabled
1: Fm+ enabled
- Bit 3 **I2C4FMP**: I2C4 Fm+
This bit enables Fm+ on I2C4.
0: Fm+ disabled
1: Fm+ enabled
- Bit 2 **I2C3FMP**: I2C3 Fm+
This bit enables Fm+ on I2C3.
0: Fm+ disabled
1: Fm+ enabled
- Bit 1 **I2C2FMP**: I2C2 Fm+
This bit enables Fm+ on I2C2.
0: Fm+ disabled
1: Fm+ enabled
- Bit 0 **I2C1FMP**: I2C1 Fm+
This bit enables Fm+ on I2C1.
0: Fm+ disabled
1: Fm+ enabled

12.3.2 SYSCFG external interrupt configuration register 1 (SYSCFG_EXTICR1)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI3[3:0]				EXTI2[3:0]				EXTI1[3:0]				EXTI0[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 0 to 3)

These bits are written by software to select the source input for the EXTI input for external interrupt / event detection.

0000: PA[x] pin

0001: PB[x] pin

0010: PC[x] pin

0011: PD[x] pin

0100: PE[x] pin

0101: PF[x] pin

0110: PG[x] pin

0111: PH[x] pin

1000: PI[x] pin

1001: PJ[x] pin

1010: PK[x] pin

Other configurations: reserved

12.3.3 SYSCFG external interrupt configuration register 2 (SYSCFG_EXTICR2)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI7[3:0]				EXTI6[3:0]				EXTI5[3:0]				EXTI4[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 4 to 7)

These bits are written by software to select the source input for the EXTI input for external interrupt / event detection.

0000: PA[x] pin

0001: PB[x] pin

0010: PC[x] pin

0011: PD[x] pin

0100: PE[x] pin

0101: PF[x] pin

0110: PG[x] pin

0111: PH[x] pin

1000: PI[x] pin

1001: PJ[x] pin

1010: PK[x] pin

Other configurations: reserved

12.3.4 SYSCFG external interrupt configuration register 3 (SYSCFG_EXTICR3)

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI11[3:0]				EXTI10[3:0]				EXTI9[3:0]				EXTI8[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 8 to 11)

These bits are written by software to select the source input for the EXTI input for external interrupt / event detection.

0000: PA[x] pin

0001: PB[x] pin

0010: PC[x] pin

0011: PD[x] pin

0100: PE[x] pin

0101: PF[x] pin

0110: PG[x] pin

0111: PH[x] pin

1000: PI[x] pin

1001: PJ[x] pin

1010: PK[x] pin

Other configurations: reserved

Note: PK[11:8] are not used

12.3.5 SYSCFG external interrupt configuration register 4 (SYSCFG_EXTICR4)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI15[3:0]				EXTI14[3:0]				EXTI13[3:0]				EXTI12[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 12 to 15)

These bits are written by software to select the source input for the EXTI input for external interrupt / event detection.

0000: PA[x] pin

0001: PB[x] pin

0010: PC[x] pin

0011: PD[x] pin

0100: PE[x] pin

0101: PF[x] pin

0110: PG[x] pin

0111: PH[x] pin

1001: PJ[x] pin

1010: PK[x] pin

Other configurations: reserved

Note: PK[15:12] are not used.

12.3.6 SYSCFG configuration register (SYSCFG_CFGR)

Address offset: 0x18

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AXI SRAML	ITCML	DTCML	SRAM1L	SRAM2L	SRAM3L	SRAM4L	Res.	BK RAML	CM7L	Res.	Res.	FLASHL	PVDL	Res.	Res.
rs	rs	rs	rs	rs	rs	rs		rs	rs			rs	rs		

Bits 31:16 Reserved, must be kept at reset value.

Bit 15 **AXISRAML**: D1 AXI-SRAM double ECC error lock bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the AXI-SRAM double ECC error flag connection to TIM1/8/15/16/17 and HRTIMER Break inputs.

0: AXI-SRAM double ECC error flag disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: AXI-SRAM double ECC error flag connected to TIM1/8/15/16/17/HRTIMER Break inputs.

Bit 14 **ITCML**: D1 ITCM double ECC error lock bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the ITCM double ECC error flag connection to TIM1/8/15/16/17 and HRTIMER Break inputs.

0: ITCM double ECC error flag disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: ITCM double ECC error flag connected to TIM1/8/15/16/17/HRTIMER Break inputs

Bit 13 **DTCML**: D1 DTCM double ECC error lock bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the DTCM double ECC error flag connection to TIM1/8/15/16/17 and HRTIMER Break inputs.

0: DTCM double ECC error flag disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: DTCM double ECC error flag connected to TIM1/8/15/16/17/HRTIMER Break inputs

Bit 12 **SRAM1L**: D2 SRAM1 double ECC error lock bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the D2 SRAM1 double ECC error flag connection to TIM1/8/15/16/17 and HRTIMER Break inputs.

0: D2 SRAM1 double ECC error flag disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: D2 SRAM1 double ECC error flag connected to TIM1/8/15/16/17/HRTIMER Break inputs

Bit 11 SRAM2L: D2 SRAM2 double ECC error lock bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the D2 SRAM2 double ECC error flag connection to TIM1/8/15/16/17 and HRTIMER Break inputs.

0: D2 SRAM2 double ECC error flag disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: D2 SRAM2 double ECC error flag connected to TIM1/8/15/16/17/HRTIMER Break inputs

Bit 10 SRAM3L: D2 SRAM3 double ECC error lock bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the D2 SRAM3 double ECC error flag connection to TIM1/8/15/16/17 and HRTIMER Break inputs.

0: D2 SRAM3 double ECC error flag disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: D2 SRAM3 double ECC error flag connected to TIM1/8/15/16/17/HRTIMER Break inputs

Bit 9 SRAM4L: D3 SRAM4 double ECC error lock bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the D3 SRAM4 double ECC error flag connection to TIM1/8/15/16/17 and HRTIMER Break inputs.

0: D3 SRAM4 double ECC error flag disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: D3 SRAM4 double ECC error flag connected to TIM1/8/15/16/17/HRTIMER Break inputs

Bit 8 Reserved, must be kept at reset value.

Bit 7 BKRAML: Backup SRAM double ECC error lock bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the Backup SRAM double ECC error flag connection to TIM1/8/15/16/17 and HRTIMER Break inputs.

0: Backup SRAM double ECC error flag disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: Backup SRAM double ECC error flag connected to TIM1/8/15/16/17/HRTIMER Break inputs

Bit 6 CM7L: Arm[®] Cortex[®]-M7 LOCKUP (HardFault) output enable bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the connection of the Arm[®] Cortex[®]-M7 LOCKUP (HardFault) output to TIM1/8/15/16/17 and HRTIMER Break inputs.

0: Arm[®] Cortex[®]-M7 LOCKUP output disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: Arm[®] Cortex[®]-M7 LOCKUP output connected to TIM1/8/15/16/17/HRTIMER Break inputs

Bits 5:4 Reserved, must be kept at reset value.

Bit 3 FLASHL: FLASH double ECC error lock bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the FLASH double ECC error flag connection to TIM1/8/15/16/17 and HRTIMER Break inputs.

0: FLASH double ECC error flag disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: FLASH double ECC error flag connected to TIM1/8/15/16/17/HRTIMER Break inputs

Bit 2 PVDL: PVD lock enable bit

This bit is set by software and cleared only by a system reset. It can be used to enable and lock the PVD connection to TIM1/8/15/16/17 and HRTIMER Break inputs, as well as the PVDE and PLS[2:0] in the PWR_CR1 register.

0: PVD signal disconnected from TIM1/8/15/16/17/HRTIMER Break inputs

1: PVD signal connected to TIM1/8/15/16/17/HRTIMER Break inputs

Bits 1:0 Reserved, must be kept at reset value.

12.3.7 SYSCFG compensation cell control/status register (SYSCFG_CCCSR)

Address offset: 0x20

Reset value: 0x0000 0000

Refer to [Section 11.3.11: I/O compensation cell](#) for a detailed description of I/O compensation mechanism.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HSLV
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	READY	Res.	Res.	Res.	Res.	Res.	Res.	CS	EN
							r							rw	rw

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 HSLV: High-speed at low-voltage

This bit is written by software to optimize the I/O speed when the product voltage is low.

This bit is active only if IO_HSLV user option bit is set. It must be used only if the product supply voltage is below 2.7 V. Setting this bit when V_{DD} is higher than 2.7 V might be destructive.

0: No I/O speed optimization

1: I/O speed optimization

Bits 15:9 Reserved, must be kept at reset value.

Bit 8 READY: Compensation cell ready flag

This bit provides the status of the compensation cell.

0: I/O compensation cell not ready

1: I/O compensation cell ready

Note: The CSI clock is required for the compensation cell to work properly. The compensation cell ready bit (READY) is not set if the CSI clock is not enabled.

Bits 7:2 Reserved, must be kept at reset value.

Bit 1 CS: Code selection

This bit selects the code to be applied for the I/O compensation cell.

0: Code from the cell (available in the SYSCFG_CCVR)

1: Code from the SYSCFG compensation cell code register (SYSCFG_CCCR)

Bit 0 EN: Enable

This bit enables the I/O compensation cell.

0: I/O compensation cell disabled

1: I/O compensation cell enabled

12.3.8 SYSCFG compensation cell value register (SYSCFG_CCVR)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCV[3:0]				NCV[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **PCV[3:0]**: PMOS compensation value

This value is provided by the cell and can be used by the CPU to compute an I/O compensation cell code for PMOS transistors. This code is applied to the I/O compensation cell when the CS bit of the SYSCFG_CCCSR is reset.

Bits 3:0 **NCV[3:0]**: NMOS compensation value

This value is provided by the cell and can be used by the CPU to compute an I/O compensation cell code for NMOS transistors. This code is applied to the I/O compensation cell when the CS bit of the SYSCFG_CCCSR is reset.

12.3.9 SYSCFG compensation cell code register (SYSCFG_CCCR)

Address offset: 0x28

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCC[3:0]				NCC[3:0]			
								rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **PCC[3:0]**: PMOS compensation code

These bits are written by software to define an I/O compensation cell code for PMOS transistors. This code is applied to the I/O compensation cell when the CS bit of the SYSCFG_CCCSR is set.

Bits 3:0 **NCC[3:0]**: NMOS compensation code

These bits are written by software to define an I/O compensation cell code for NMOS transistors. This code is applied to the I/O compensation cell when the CS bit of the SYSCFG_CCCSR is set.

12.3.10 SYSCFG power control register (SYSCFG_PWRCR)

Address Offset: 0x2C

Reset Value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ODEN
															rw

Bits 31:1 Reserved, must be kept at reset value.

- Bit 0 **ODEN**: Overdrive enable, this bit allows to activate the LDO regulator overdrive mode.
 This bit must be written only in VOS1 voltage scaling mode.
 0: Overdrive mode disabled
 1: Overdrive mode enabled (the LDO generates VOS0 for V_{CORE})

Note: VOS0 must be activated only in VOS1 mode. It must be disabled by software before entering low-power mode (execution of WFE/WFI instruction).

12.3.11 SYSCFG package register (SYSCFG_PKGR)

Address offset: 0x124

Reset value: 0x0000 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PKG[3:0]			
												r	r	r	r

Bits 31:4 Reserved, must be kept at reset value.

- Bits 3:0 **PKG[3:0]**: Package
 These bits indicate the device package.
 0000: LQFP100
 0010: TQFP144
 0101: TQFP176/UFPGA176
 1000: LQFP208/TFBGA240
 Other configurations: all pads enabled

12.3.12 SYSCFG user register 0 (SYSCFG_UR0)

Address offset: 0x300

Reset value: 0x00XX 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RDP[7:0]							
								r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BKS
															r

Bits 31:24 Reserved, must be kept at reset value.

Bits 23:16 **RDP[7:0]**: Readout protection

These bits indicate the readout protection level.

0xAA: Level 0 (no protection)

0xCC: Level 2 (Flash memory readout protected, full debug features, boot from SRAM and boundary scan disabled)

Other configurations: Level 1 (Flash memory readout protected, limited debug features and boundary scan enabled)

Bits 15:1 Reserved, must be kept at reset value.

Bit 0 **BKS**: Bank Swap

This bit indicates Flash memory bank mapping.

0: Flash memory bank addresses are inverted

1: Flash memory banks are mapped to their original addresses

12.3.13 SYSCFG user register 2 (SYSCFG_UR2)

Address offset: 0x308

Reset value: 0xFFFF 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BOOT_ADD0[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BORH[1:0]	
														r	r

Bits 31:16 **BOOT_ADD0[15:0]**: Boot Address 0

These bits define the MSB of the core boot address when BOOT pin is low.

Bits 15:2 Reserved, must be kept at reset value.

Bits 1:0 **BORH[1:0]**: BOR_LVL Brownout Reset Threshold Level

These bits indicate the Brownout reset high level.

0x11: BOR Level 3

0x10: BOR Level 2

0x01: BOR Level 1

0x00: BOR OFF (Level 0)

12.3.14 SYSCFG user register 3 (SYSCFG_UR3)

Address offset: 0x30C

Reset value: 0xFFFF XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BOOT_ADD1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **BOOT_ADD1[15:0]**: Boot Address 1

These bits define the MSB of the core boot address when BOOT pin is high.

12.3.15 SYSCFG user register 4 (SYSCFG_UR4)

Address offset: 0x310

Reset value: 0x000X XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MEPAD_1
															r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **MEPAD_1**: Mass Erase Protected Area Disabled for bank 1

This bit indicates if the flash protected area (Bank 1) is affected by a mass erase.

0: When a mass erase occurs the protected area is erased

1: When a mass erase occurs the protected area is not erased

Bits 15:0 Reserved, must be kept at reset value.

12.3.16 SYSCFG user register 5 (SYSCFG_UR5)

Address offset: 0x314

Reset value: 0x00XX 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	WRPS_1[7:0]							
								r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MESAD_1
															r

Bits 31:24 Reserved, must be kept at reset value.

Bits 23:16 **WRPN_1[7:0]**: Write protection for flash bank 1

WRPN[i] bit indicates if the sector i of the Flash memory bank 1 is protected.

0: Write protection is active on sector i

1: Write protection is not active on sector i

Bits 15:1 Reserved, must be kept at reset value.

Bit 0 **MESAD_1**: Mass erase secured area disabled for bank 1

This bit indicates if the flash secured area (bank 1) is affected by a mass erase.

0: When a mass erase occurs the secured area is erased

1: When a mass erase occurs the secured area is not erased

12.3.17 SYSCFG user register 6 (SYSCFG_UR6)

Address offset: 0x318

Reset value: 0x0XXX 0XXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	PA_END_1[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	PA_BEG_1[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:28 Reserved, must be kept at reset value.

Bits 23:16 **PA_END_1[11:0]**: Protected area end address for bank 1

End address for bank 1 protected area.

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:0 **PA_BEG_1[11:0]**: Protected area start address for bank 1

Start address for bank 1 protected area.

12.3.18 SYSCFG user register 7 (SYSCFG_UR7)

Address offset: 0x31C

Reset value: 0x0XXX 0XXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	SA_END_1[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	SA_BEG_1[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:28 Reserved, must be kept at reset value.

Bits 23:16 **SA_END_1[11:0]**: Secured area end address for bank 1
End address for bank 1 secured area.

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:0 **SA_BEG_1[11:0]**: Secured area start address for bank 1
Start address for bank 1 secured area.

12.3.19 SYSCFG user register 8 (SYSCFG_UR8)

Address offset: 0x320

Reset value: 0x000X 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MESAD_2
															r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MEPAD_2
															r

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **MESAD_2**: Mass erase secured area disabled for bank 2

This bit indicates if the Flash memory secured area (Bank 2) is affected by a mass erase.

0: When a mass erase occurs the secured area is erased

1: When a mass erase occurs the secured area is not erased

Bits 15:1 Reserved, must be kept at reset value.

Bit 0 **MEPAD_2**: Mass erase protected area disabled for bank 2

This bit indicates if the Flash memory protected area (Bank 2) is affected by a mass erase.

0: When a mass erase occurs the protected area is erased

1: When a mass erase occurs the protected area is not erased

12.3.20 SYSCFG user register 9 (SYSCFG_UR9)

Address offset: 0x324

Reset value: 0x0XXX 00XX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	PA_BEG_2[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	WRPS_2[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:16 **PA_BEG_2[11:0]**: Protected area start address for bank 2
 Start address for bank 2 protected area.

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:0 **WRPN_2[7:0]**: Write protection for flash bank 2
 WRPN[i] bit indicates if the sector i of the Flash memory bank 2 is protected.
 0: Write protection is active on sector i
 1: Write protection is not active on sector i

12.3.21 SYSCFG user register 10 (SYSCFG_UR10)

Address offset: 0x328

Reset value: 0x0XXX 0XXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	SA_BEG_2[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	PA_END_2[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:16 **SA_BEG_2[11:0]**: Secured area start address for bank 2
 Start address for bank 2 secured area.

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:0 **PA_END_2[11:0]**: Protected area end address for bank 2
 End address for bank 2 protected area.

12.3.22 SYSCFG user register 11 (SYSCFG_UR11)

Address offset: 0x32C

Reset value: 0x000X 0XXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IWDG1M
															r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	SA_END_2[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **IWDG1M**: Independent Watchdog 1 mode

This bit indicates the control mode of the Independent Watchdog 1 (IWDG1).

0: IWDG1 controlled by hardware

1: IWDG1 controlled by software

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:0 **SA_END_2[11:0]**: Secured area end address for bank 2

End address for bank 2 secured area.

12.3.23 SYSCFG user register 12 (SYSCFG_UR12)

Address offset: 0x330

Reset value: 0x000X 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECURE
															r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
															r

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **SECURE**: Secure mode

This bit indicates the Secure mode status.

0: Secure mode disabled

1: Secure mode enabled

Bits 15:0 Reserved, must be kept at reset value.

12.3.24 SYSCFG user register 13 (SYSCFG_UR13)

Address offset: 0x334

Reset value: 0x000X 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	D1SBRST
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDRS[1:0]	
														r	r

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **D1SBRST**: D1 Standby reset

This bit indicates if a reset is generated when D1 domain enters DStandby mode.

0: A reset is generated by entering D1 Standby mode

1: D1 Standby mode is entered without reset generation

Bits 15:2 Reserved, must be kept at reset value.

Bits 1:0 **SDRS[1:0]**: Secured DTCM RAM Size

These bits indicates the size of the secured DTCM RAM.

00: 2 Kbytes

01: 4 Kbytes

10: 8 Kbytes

11: 16 Kbytes

12.3.25 SYSCFG user register 14 (SYSCFG_UR14)

Address offset: 0x338

Reset value: 0x000X 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	D1STPRST
															rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:1 Reserved, must be kept at reset value.

Bit 0 **D1STPRST**: D1 Stop Reset

This bit indicates if a reset is generated when D1 domain enters in DStop mode.

0: A reset is generated entering D1 Stop mode

1: D1 Stop mode is entered without reset generation

12.3.26 SYSCFG user register 15 (SYSCFG_UR15)

Address offset: 0x33C

Reset value: 0x000X 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FZIWDS TB
															r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **FZIWDS****TB**: Freeze independent watchdog in Standby mode

This bit indicates if the independent watchdog is frozen in Standby mode.

0: Independent Watchdog frozen in Standby mode

1: Independent Watchdog running in Standby mode

Bits 15:0 Reserved, must be kept at reset value.

12.3.27 SYSCFG user register 16 (SYSCFG_UR16)

Address offset: 0x340

Reset value: 0x000X 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PKP
															r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FZIWDG STP
															r

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **PKP**: Private key programmed

This bit indicates if the device private key is programmed.

0: Private key not programmed

1: Private key programmed

Bits 15:1 Reserved, must be kept at reset value.

Bit 0 **FZIWDGSTP**: Freeze independent watchdog in Stop mode

This bit indicates if the independent watchdog is frozen in Stop mode.

0: Independent Watchdog frozen in Stop mode

1: Independent Watchdog running in Stop mode

12.3.28 SYSCFG user register 17 (SYSCFG_UR17)

Address offset: 0x344

Reset value: 0x0000 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IO_HSLV
															r

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **IO_HSLV**: I/O high speed / low voltage

This bit indicates that the IOHSLV option bit is set.

0: Product is working on the full voltage range

1: Product is working below 2.7 V

12.3.29 SYSCFG register maps

The following table gives the SYSCFG register map and the reset values.

Table 94. SYSCFG register map and reset values

Offset	Register name	31		30		29		28		27		26		25		24		23		22		21		20		19		18		17		16		15		14		13		12		11		10		9		8		7		6		5		4		3		2		1		0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
0x00	Reserved	Reserved																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
0x04	SYSCFG_PMCRR	Res.		Res.		Res.		Res.		Res.		PC3SO		PC2SO		PA1SO		PA0SO				EPIS[2:0]					Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.		Res.	</

Table 94. SYSCFG register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
0x304	Reserved	Reserved																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
0x308	SYSCFG_UR2	BOOT_ADD0[15:0]																Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Table 94. SYSCFG register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x338	SYSCFG_UR14	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	D1STRST
	Reset value																X																X
0x33C	SYSCFG_UR15	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FZWDGSTB	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.
	Reset value																X																X
0x340	SYSCFG_UR16	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PKP	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	FZWDGST
	Reset value																X																X
0x344	SYSCFG_UR17	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	RES.	IO_HSLV
	Reset value																																X

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.